

Retail Customer Retention Analytics – TESCO

Power BI Project Report

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Course: Data Analytics

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PROJECT INTRODUCTION

This project focuses on analyzing customer retention for Tesco using Power BI. The aim is to understand customer churn behavior, loyalty program impact, store performance, and customer lifetime value to generate actionable business insights.

PROBLEM STATEMENT

Customer churn directly impacts business revenue and growth. Tesco operates across multiple locations with large volumes of customer and transaction data, but lacks unified analytics to identify churn risks, high-value customers, and the effectiveness of loyalty programs.

DATA DESCRIPTION

Datasets used:

- Customer Demographics
- Customer Transactions
- Loyalty Program Data
- Store Location Data
- Churn Label Dataset.

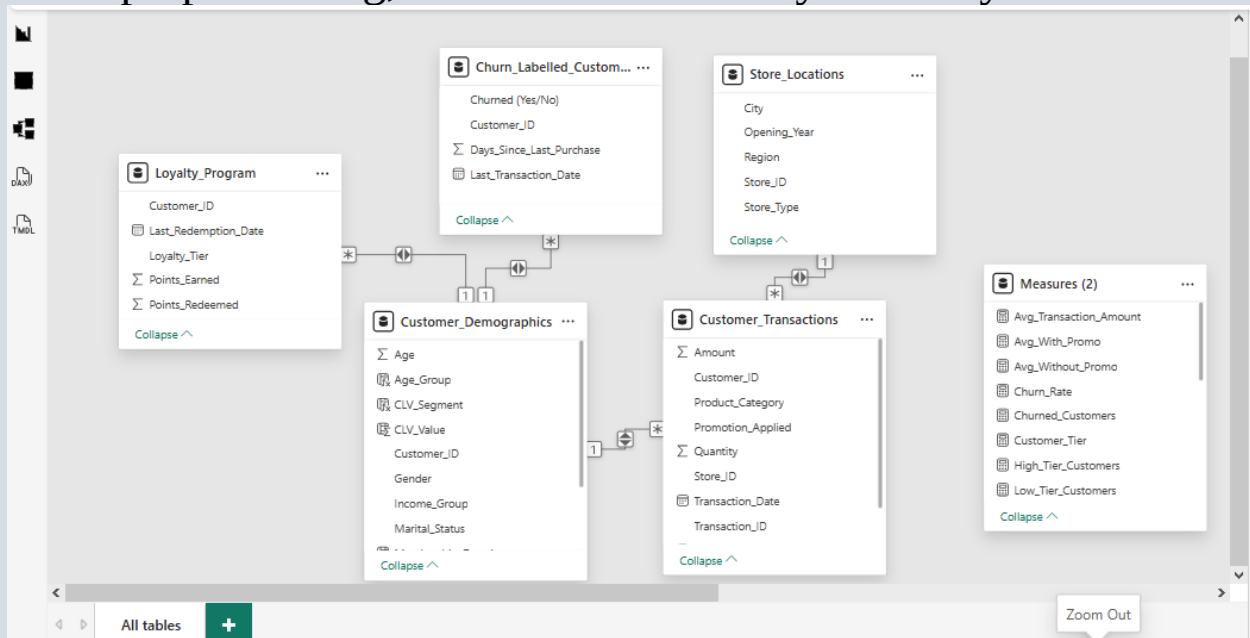
TASK 1 — Data Cleaning & Modeling

In this task, data was prepared for analysis using Power Query and data modeling techniques. Data cleaning ensured consistency and accuracy before building the dashboard.

Steps performed:

- Removed duplicate records
- Corrected data types
- Standardized date formats
- Handled missing values
- Created relationships between tables
- Created calculated columns like Membership Duration, Transaction Year, and Month

After preprocessing, the dataset was ready for analysis.



Power BI Tesco Project

Home Transform Add Column View Tools Help

Close & Apply ~ New Source ~ Recent Sources ~ Enter Data ~ Data source settings ~ Manage Parameters ~ Export query results ~ Refresh Preview ~ Advanced Editor ~ Choose Columns ~ Remove Columns ~ Keep Rows ~ Remove Rows ~ Sort ~ Split Column ~ Group By ~ Data Type: Text ~ Use First Row as Headers ~ Replace Values ~ Merge ~ Append ~ Combine ~ Correlate

Queries [7]

Customer_Demographics

Customer_Transactions

Store_Locations

Loyalty_Program

Churn_Labelled_Custom...

Measures (2)

CLV measure

Table.SelectRows(*Reordered Columns*, each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"", ""})))

	Customer_ID	Age	Gender	Membership_Since	Marital_Status	Region
1	C1000		50 Male	11/1/2020	Single	London
2	C1001		18 Female	7/5/2021	Divorced	London
3	C1002		36 Male	8/18/2021	Single	Birmingham
4	C1003		19 Male	2/1/2024	Married	Leeds
5	C1004		70 Male	10/15/2020	Married	Leeds
6	C1005		61 Female	11/19/2020	Single	Liverpool
7	C1006		43 Male	7/13/2021	Divorced	London
8	C1007		49 Male	3/16/2023	Married	London
9	C1008		23 Male	6/14/2022	Married	Manchester
10	C1009		49 Female	6/6/2024	Married	Liverpool
11	C1010		21 Male	5/3/2023	Single	Manchester
12	C1011		28 Male	7/7/2021	Married	Manchester
13	C1012		34 Male	2/27/2023	Divorced	Manchester
14	C1013		55 Male	4/27/2023	Married	Birmingham
15	C1014		41 Female	1/20/2024	Married	Birmingham
16	C1015		22 Male	11/15/2023	Divorced	Manchester
17	C1016		69 Female	5/18/2023	Married	Liverpool
18	C1017		51 Female	7/11/2023	Single	London
19	C1018		23 Female	3/21/2024	Single	Liverpool
20						

7 COLUMNS, 300 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 7:44 PM

Query Settings

Customer_Demographics

PROPERTIES

APPLIED STEPS

Source

Promoted Headers

Removed Duplicates

Changed Type

Filtered Rows

Reordered Columns

Removed Blank Rows

TASK 2 — Churn & Retention Metrics

This task focused on analyzing customer churn and retention patterns. Key performance indicators such as churn rate and repeat purchase rate were calculated using DAX measures.

Measures used:

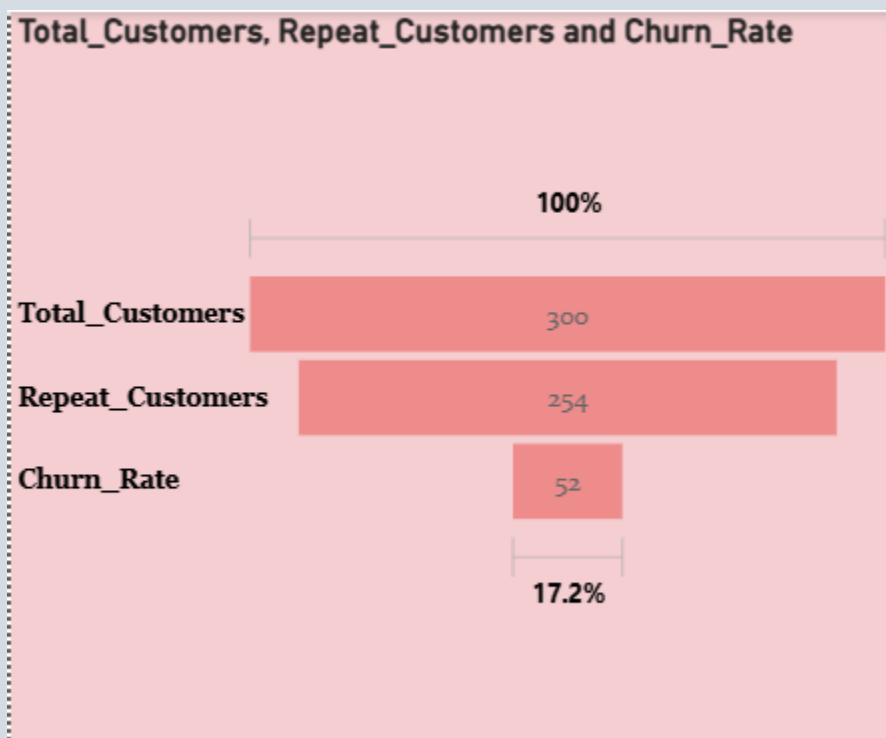
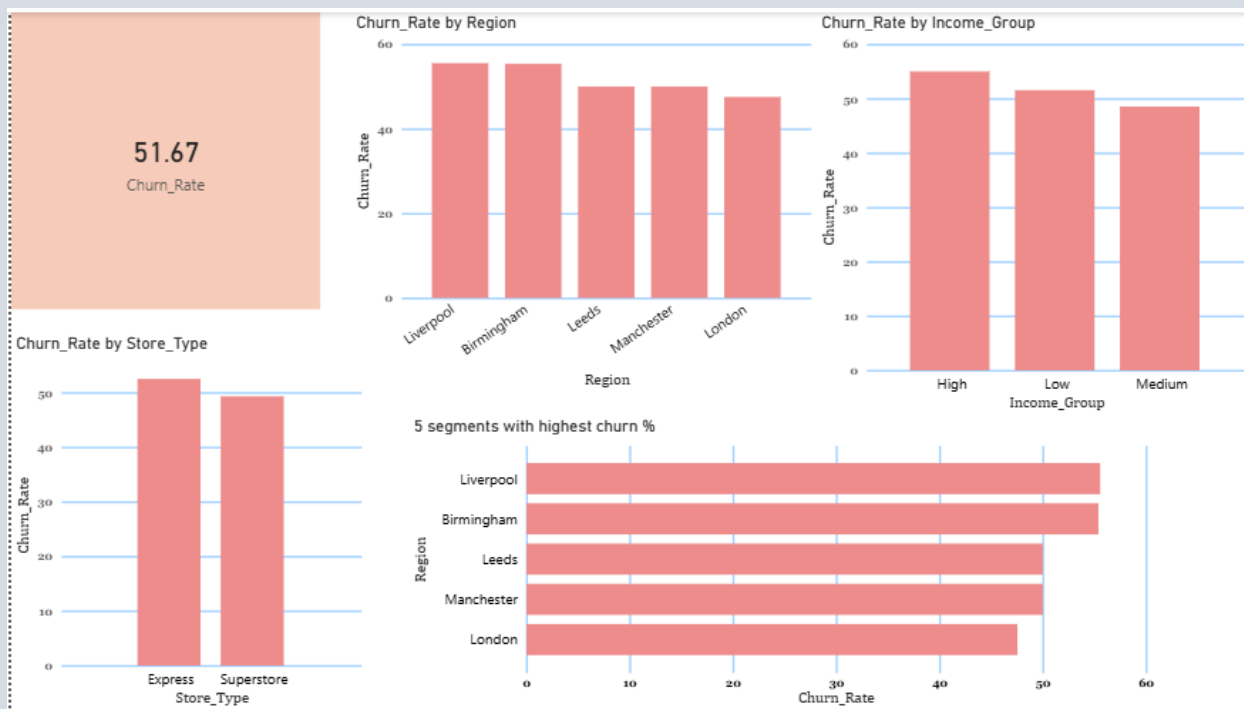
Churn Rate =

$(\text{Distinct Churned Customers} / \text{Total Customers}) \times 100$

Repeat Rate =

$(\text{Customers with multiple purchases} / \text{Total Customers})$

Churn trends were analyzed by region, income group, and store type.



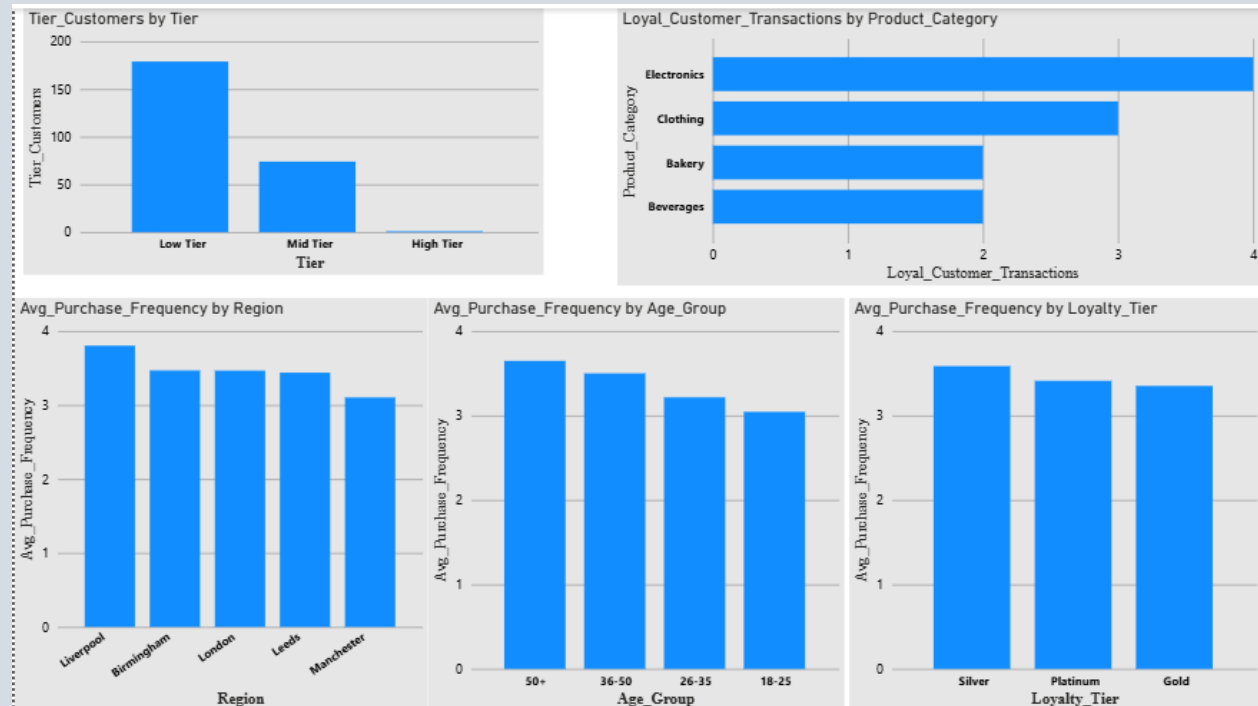
TASK 3 — Repeat Purchase Analysis

In this task, customer purchase frequency was analyzed to identify loyal customers. Customers were segmented into different tiers based on purchase count.

Segments created:

- Low Tier (2–4 purchases)
- Mid Tier (5–10 purchases)
- High Tier (11+ purchases)

This analysis helped identify repeat customers contributing significantly to revenue.



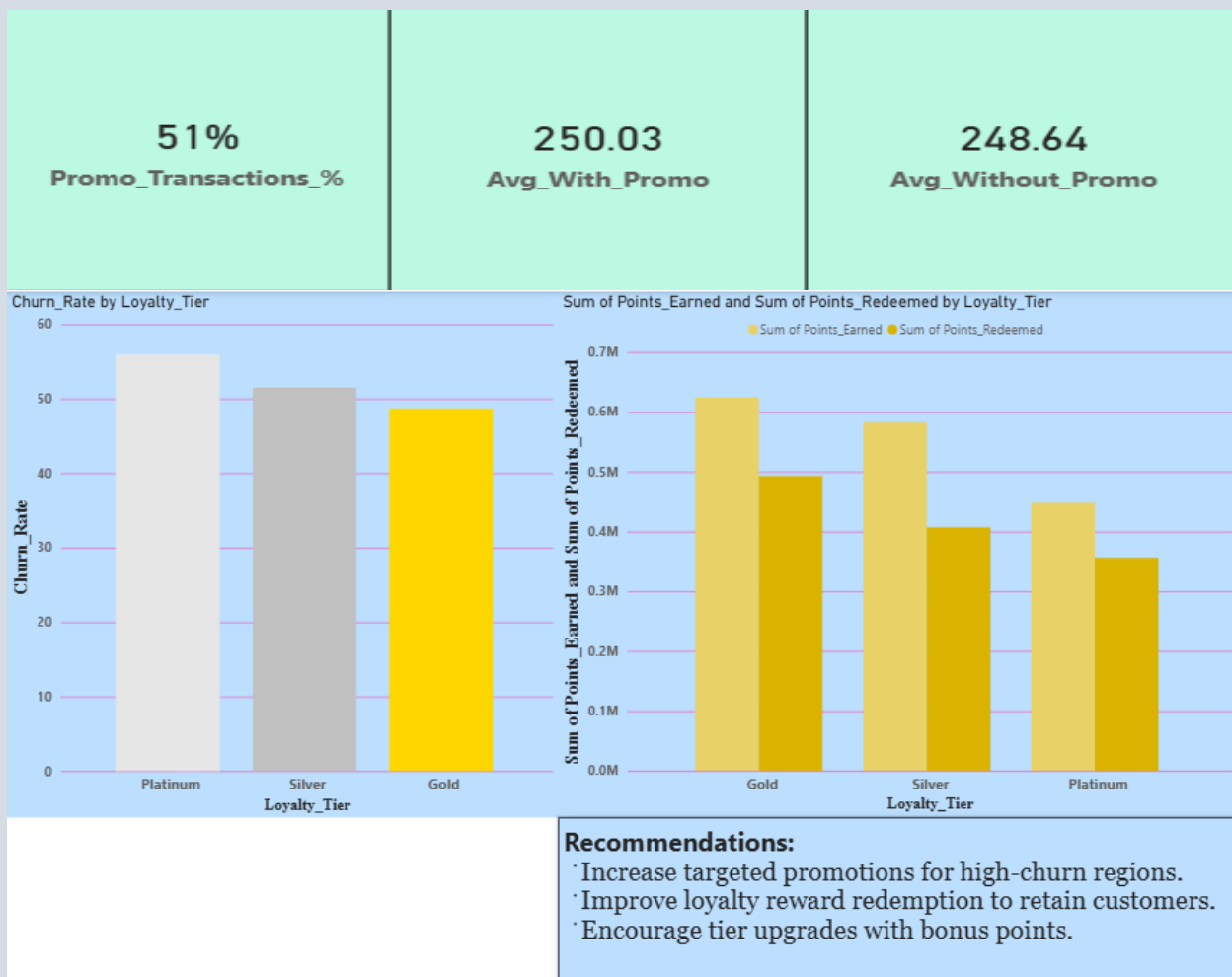
TASK 4 — Promotion & Loyalty Impact

This task analyzed the impact of promotions and loyalty programs on customer retention and spending behavior.

Measures created:

Promotion Percentage =
(Promotion Transactions / Total Transactions)

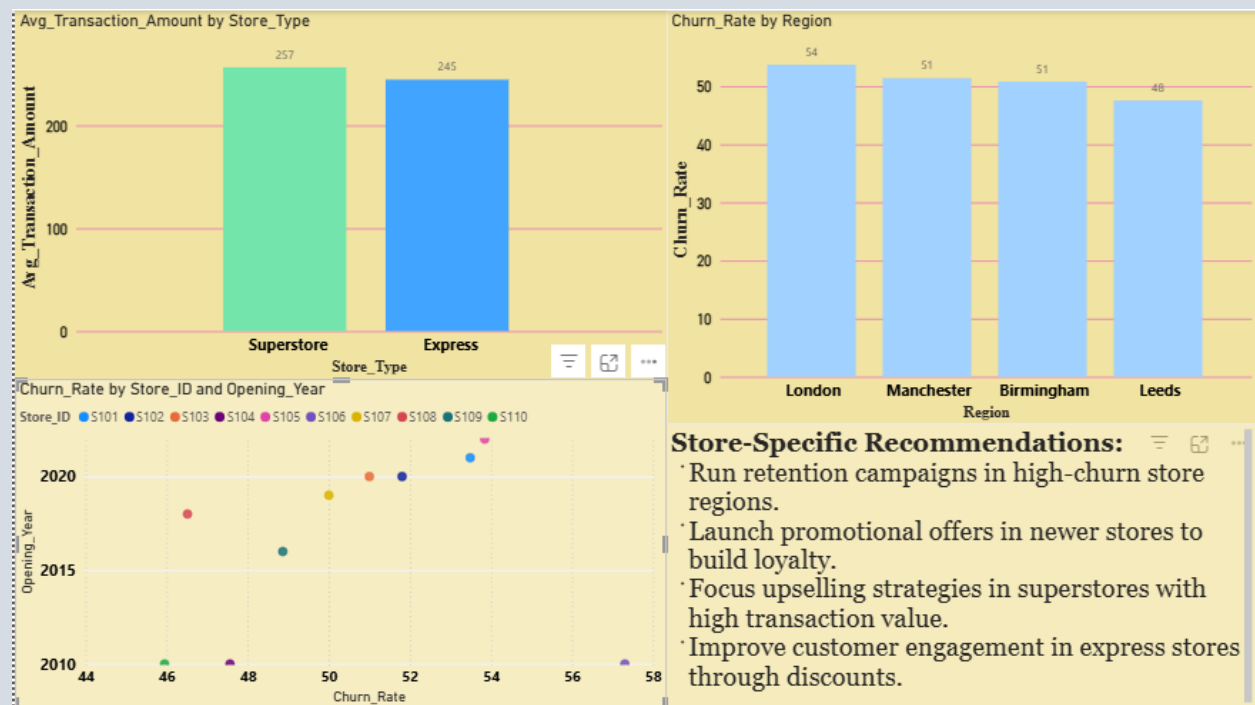
Average purchase with promotion and without promotion was compared. Loyalty tier analysis helped understand customer engagement.



TASK 5 — Store Performance Analysis:

This task analyzed how store characteristics influence customer retention. Average transaction value, churn rate by region, and store opening year impact were studied.

This helped identify regions and store types requiring targeted strategies.



TASK 6 — Customer Lifetime Value (CLV) Analysis

Customer Lifetime Value (CLV) was calculated to identify high-value customers.

Formula:

$$\text{CLV} = \text{Total Amount Spent} / \text{Membership Duration}$$

Customers were segmented into low, medium, and high CLV groups. This helped prioritize retention strategies.



Measure used for the Project :-

Measures (2)

☐ Avg_Transaction_Amount
☐ Avg_With_Promo
☐ Avg_Without_Promo
☐ Churn_Rate
☐ Churned_Customers
☐ Customer_Tier
☐ High_Tier_Customers
☐ Low_Tier_Customers
☐ Loyal_Customer_Transactions
☐ Membership_Duration(Years)
☐ Mid_Tier_Customers
☐ Promo_Transactions_%
☐ Promo_Transactions_Percent
☐ Purchase_Count
☐ Repeat_Rate
☐ Repeat_Customers
☐ Repeat_Segment_Count
☐ Tier_Customers

☐ Total_Amount_Spent
☐ Total_Customers
☐ Total_Orders
☐ Total_Transactions
> ☐ Churn_Labelled_Customers
> ☐ CLV measure

☐ Avg_Purchase_Frequency
☐ CLV
☐ CLV_25th
☐ CLV_75th
☐ CLV_Category
☐ CLV_Segment
☐ Column1
☐ High_CLV_Customers
☐ Low_CLV_Customers
☐ Medium_CLV_Customers

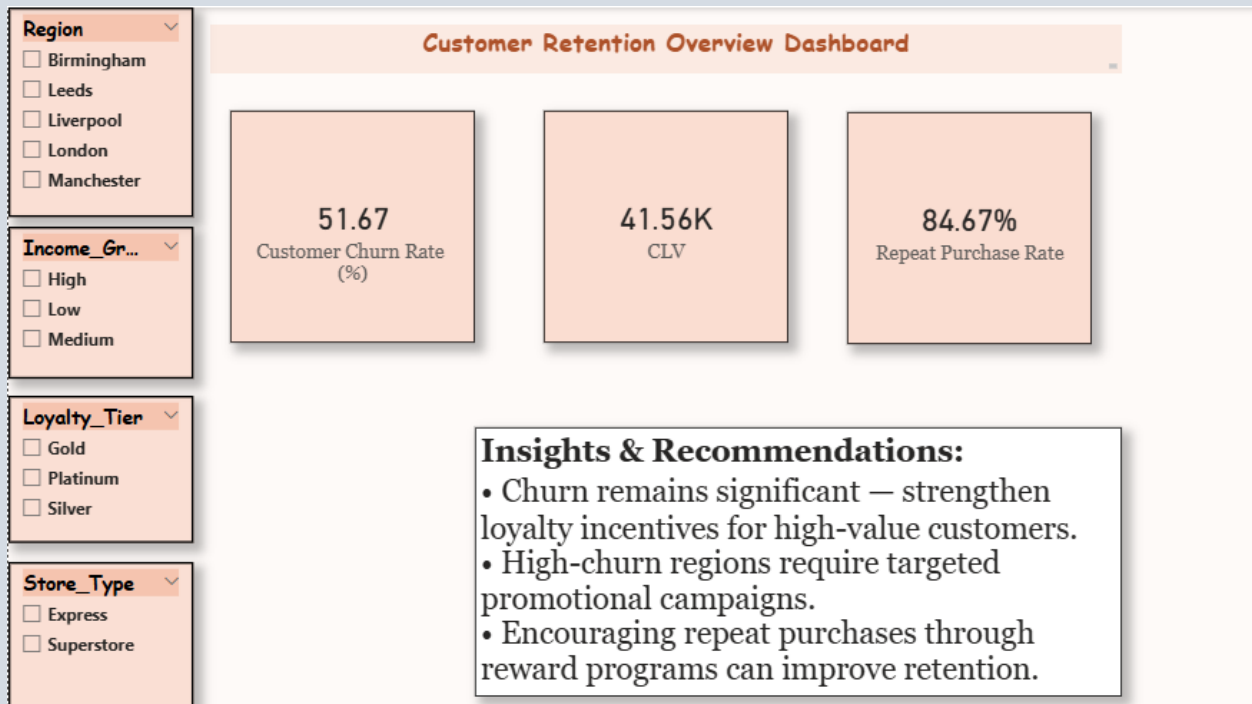
> ☐ Customer_Demographics
> ☐ Customer_Tier
> ☐ Customer_Transactions

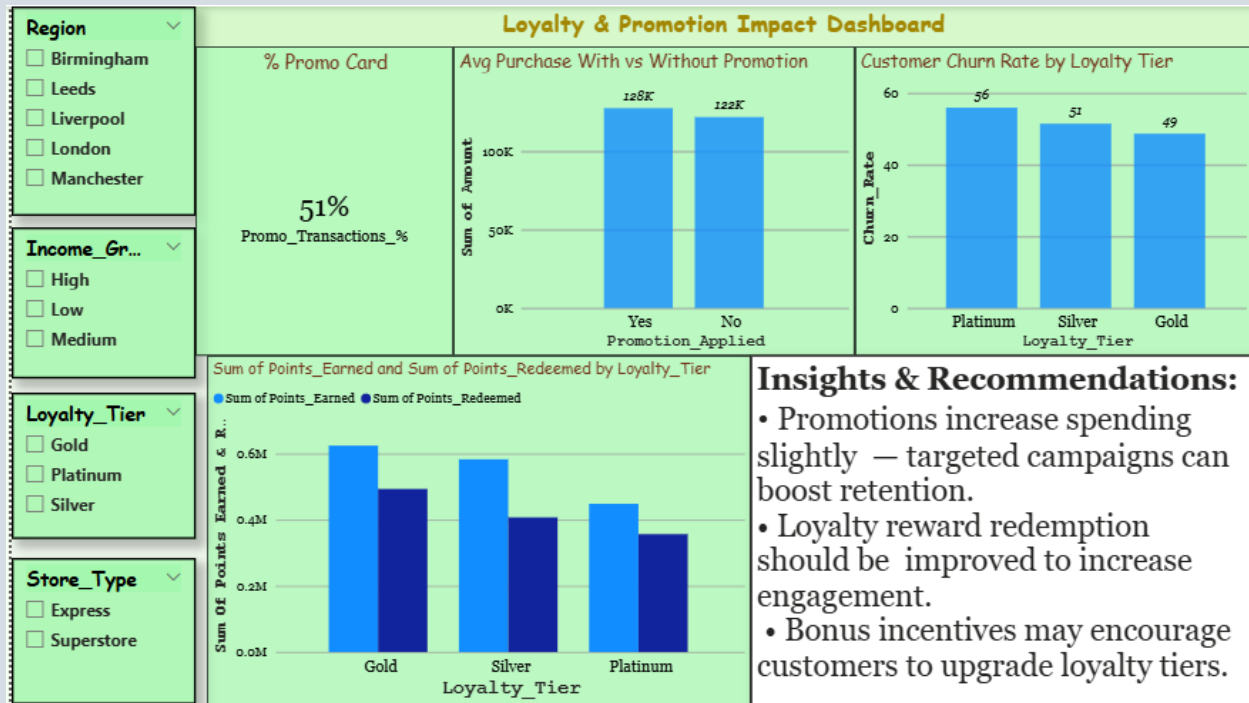
TASK 7 — Final Dashboard & Insights

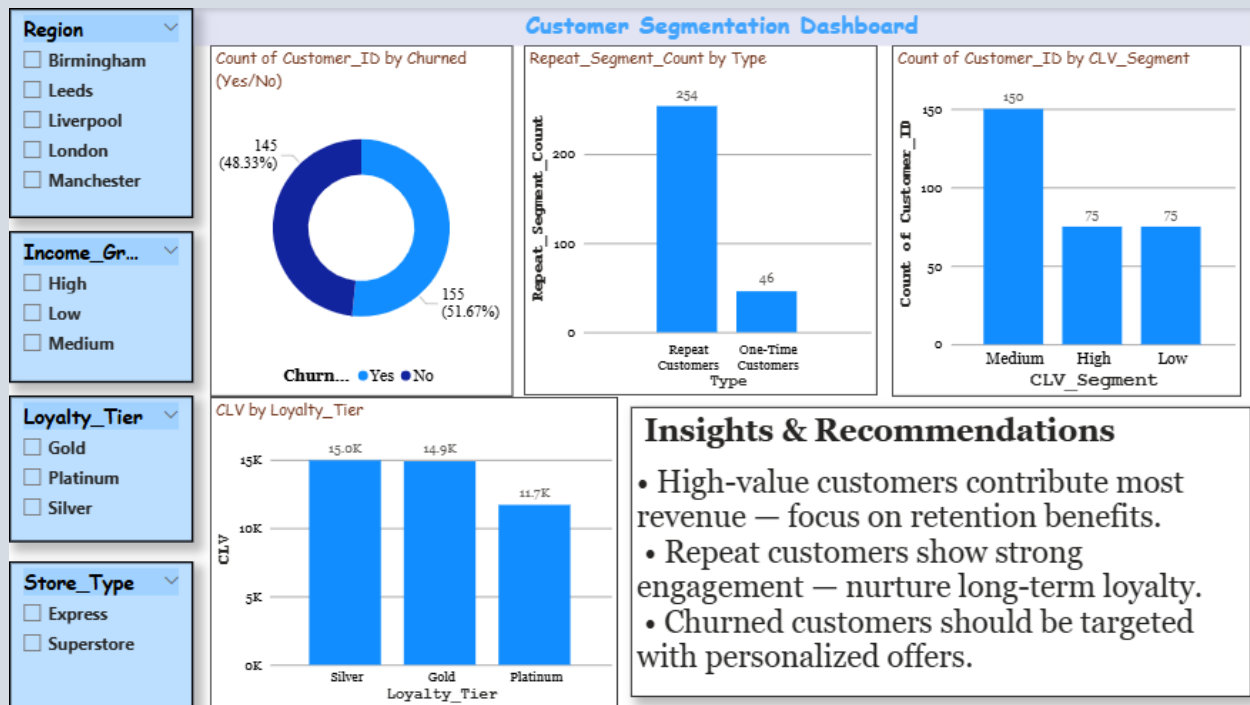
A multi-page Power BI dashboard was created to present insights clearly. It includes:

- Overview KPI Page
- Loyalty & Promotion Impact Page
- Store & Region Insights Page
- Customer Segmentation Page

Interactive slicers were added for dynamic filtering.







Executive Summary & Recommendations :-

Executive Summary & Recommendations

Key Insights:

- High-value customers contribute the majority of revenue means retention should focus on them.
- Certain regions show higher churn and require targeted engagement.
- Loyalty programs positively impact customer retention.

Recommendations:

- Strengthen loyalty incentives for high-value customers.
- Target high-churn regions with personalized promotions.
- Encourage repeat purchases through reward programs.

Conclusion

This project successfully demonstrates how Power BI can be used to analyze customer retention for Tesco. The dashboard highlights churn patterns, loyalty impact, promotion effectiveness, store performance, and customer lifetime value.

The analysis shows that high-value customers contribute significantly to revenue, while certain regions exhibit higher churn risk. Loyalty programs and targeted promotions positively influence retention.

Overall, the dashboard provides actionable insights that can support data-driven retention strategies and improve long-term customer loyalty.

Project Video Link:

<https://www.loom.com/share/7c6befe167034254ad299b3c042843ad>